

Lab Goal : This lab was designed to teach you more about graphs, mazes, and recursion.

Lab Description : A maze is a two dimensional structure. All of the mazes will have the entrance point at the top left corner. The exit point will be the bottom right corner.

You must write a program to determine if it is possible to go from the entrance point to the exit point.

Helpful Hints / Assumptions: All input characters will be '#'s and '.'s, representing walls and open spaces respectively. You can travel through open spaces '.'s but not walls '#'s.

Sample Data :

```
5
..###
....#
###.#
#....
###..
2
..
..
3
...
###
...
9
..#####
....#...#
#.###.###
#.....#
#.###.#.#
#.###.#.#
#.###.#.#
#.###.#.#
#####..
9
..#####
....#...#
#.###.###
#.....#
#.###.#.#
#.###.#.#
#.###.#.#
#####..
```

Files Needed ::

```
//OOP Lab - recursion
Maze.java
MazeRunner.java

//OOP Lab – queue
MazeQ.java
MazeQRunner.java

//non-OOP Lab
Maze_Simple.java

//data file needed for all
maze_simple.dat
```

algorithm help

```
if ( r and c are in bounds and current spot is a 1 )
    mark spot as visited
    if we are at the exit
        mark exit found as true
    else
        4 recursive calls up down left right
```

Sample Output :

```
. . # # #  
. . . . #  
# # # . #  
# . . . .  
# # # . .
```

true

```
. .  
. .
```

true

```
. . .  
# # #  
. . .
```

false

```
. . # # # # # # #  
. . . . # . . . #  
# . # # # . # # #  
# . . . . . . . #  
# . # # # . # . #  
# . # . . . # . #  
# . # . # # # . #  
# . # . # . . . #  
# # # # # # # . .
```

true

```
. . # # # # # # #  
. . . . # . . . #  
# . # # # . # # #  
# . . . . . . . #  
# . # # # . # . #  
# . # . . . # # #  
# . # . # # # . #  
# . # . # . . . #  
# # # # # # # . .
```

false